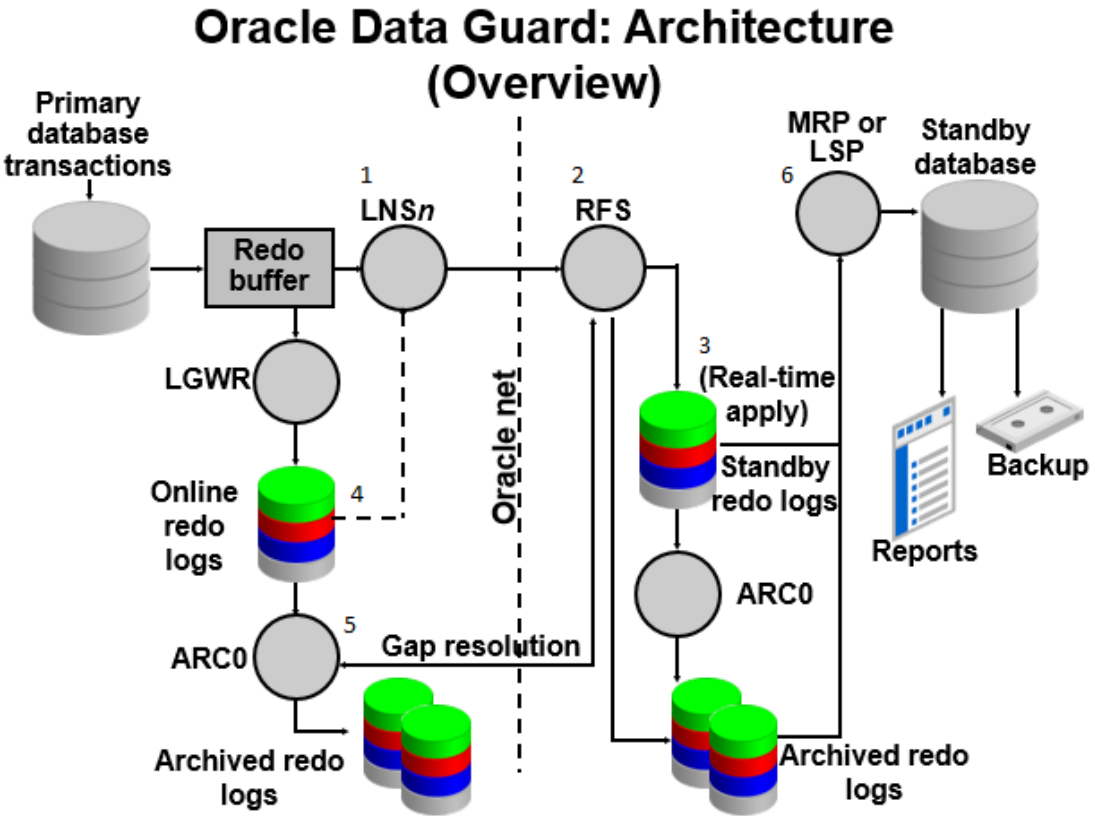


Oracle 19c Data Guard & Data Guard Broker Configuration

Complete Guide

(Step-by-Step Implementation)

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1. Introduction

Oracle Data Guard is Oracle's enterprise-level disaster recovery and high availability (HA) solution. It provides real-time data protection, automated failover/switchover capabilities, and efficient replication between primary and standby databases.

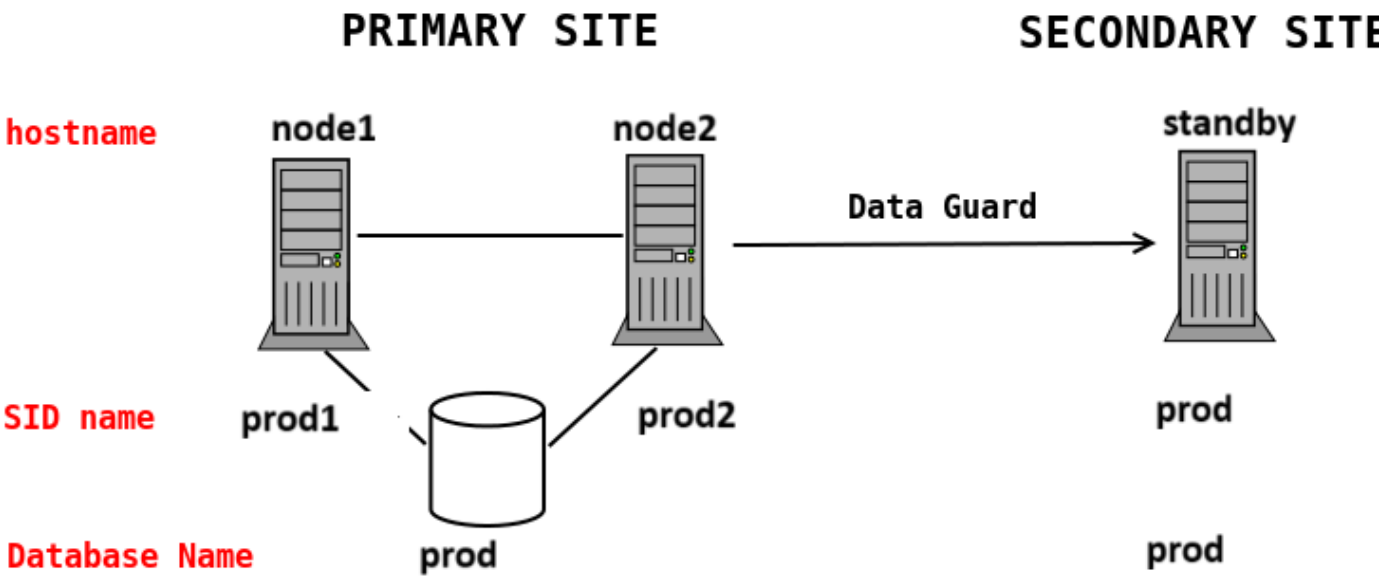
In modern enterprise environments, where business continuity is critical, implementing Data Guard between a highly available **2-Node RAC (Real Application Clusters)** primary and a **standalone physical standby** ensures:

- **High Availability (HA)**
- **Disaster Recovery (DR)**
- **Zero/Minimal Data Loss**
- **Automatic log transport and apply**
- **Seamless role transitions**

This document explains the **complete end-to-end implementation** of Oracle Data Guard and Data Guard Broker between:

- A **2-node RAC Primary CDB** running on ASM
- A **Standalone Physical Standby CDB** running on ASM

This guide includes environment details, prerequisites, configurations, duplicate process, redo apply, verification, and broker setup.



2. Architecture Overview

2.1 Primary (RAC) Configuration

Component	Details
Nodes	node1, node2
Cluster	Oracle Grid Infrastructure
Database	Container Database (CDB): **prod**
Instances	prod1, prod2
PDB	prods
DB Unique Name	PROD
Storage	ASM (+DATA, +RECO, +OCRVD)
Role	Primary
Version	Oracle 19.16
SCAN	rac-scan.localdomain

2.2 Standby (Standalone) Configuration

Component	Details
Server	standby.localdomain
Database	Container Database (CDB): **prod**
PDB	prods
Instance	prod
DB Unique Name	PRODSTBY
Storage	ASM (+DATA, +RECO)
Role	Physical Standby
Version	Oracle 19.16

3. Objectives of This Implementation

- Configure a **Physical Standby Database** for a 2-Node RAC primary.
- Enable **real-time redo transport & apply**.
- Ensure **Data Guard health and synchronization**.
- Implement **Data Guard Broker** for simplified management.
- Test and document **Switchover** and **Failover** readiness.
- Ensure **PDB-level replication** in a CDB environment.

---

4. Prerequisites

4.1 Database & OS Requirements

- Identical Oracle version (19.16) on both servers.
- ASM disk groups available on primary and standby.
- Sufficient network connectivity (ports 1521 open).

4.2 Configuration Requirements

- Primary in **ARCHIVELOG** and **FORCE LOGGING** mode.
- Oracle Password File copied to standby.
- tnsnames.ora and listener.ora configured correctly.
- DG-related initialization parameters set.
- SRLs (Standby Redo Logs) added on PRIMARY.

---

5. Primary Database Configuration

5.1 Make sure primary is in archivelog mode :

```
[oracle@node1 ~]$ . oraenv
ORACLE_SID = [prod] ? prod1
The Oracle base has been changed from /home/oracle to /u01/app/oracle
[oracle@node1 ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Nov 27 13:37:58 2025
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle.  All rights reserved.


Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL>
```

```
SQL> archive log list;
Database log mode              Archive Mode
Automatic archival             Enabled
Archive destination            USE_DB_RECOVERY_FILE_DEST
Oldest online log sequence     12
Next log sequence to archive   13
Current log sequence           13
SQL>
```

5.2 Check FORCE LOGGING is enabled :

```
SQL> select force_logging from v$database;

FORCE_LOGGING
---
NO

SQL> alter database force logging;

Database altered.

SQL> select force_logging from v$database;

FORCE_LOGGING
---
YES
```

---

6. Initialization Parameters

6.1 Set Primary Database Initialization Parameters :

```
SQL> ALTER SYSTEM SET LOG_ARCHIVE_CONFIG='DG_CONFIG=(PROD,PRODSTBY)' SCOPE=BOTH SID='*';

System altered.

SQL> ALTER SYSTEM SET LOG_ARCHIVE_DEST_1='LOCATION=USE_DB_RECOVERY_FILE_DEST
     VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=PROD' SCOPE=BOTH SID='*';

System altered.

SQL> ALTER SYSTEM SET LOG_ARCHIVE_DEST_2="SERVICE=PRODSTBY ASYNC
     VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=PRODSTBY" SCOPE=BOTH SID='*';

System altered.

SQL> ALTER SYSTEM SET LOG_ARCHIVE_DEST_STATE_1=ENABLE SCOPE=BOTH SID='*';

System altered.

SQL> ALTER SYSTEM SET LOG_ARCHIVE_DEST_STATE_2=ENABLE SCOPE=BOTH SID='*';

System altered.

SQL> ALTER SYSTEM SET FAL_SERVER='PRODSTBY' SCOPE=BOTH SID='*';

System altered.
```

```
SQL> ALTER SYSTEM SET FAL_CLIENT='PROD' SCOPE=BOTH SID='*';

System altered.

SQL> ALTER SYSTEM SET STANDBY_FILE_MANAGEMENT=AUTO SCOPE=BOTH SID='*';

System altered.

SQL> alter system set remote_login_passwordfile=exclusive scope=spfile sid='*';

System altered.

SQL>


SQL> show parameter log_archive

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
log_archive_config                   string      DG_CONFIG=(PROD,PRODSTBY)
log_archive_dest                     string
log_archive_dest_1                   string      LOCATION=USE_DB_RECOVERY_FILE_
                                                 DEST VALID_FOR=(ALL_LOGFILES,A
                                                 LL_ROLES) DB_UNIQUE_NAME=PROD
log_archive_dest_10                  string
log_archive_dest_11                  string
log_archive_dest_12                  string
log_archive_dest_13                  string
log_archive_dest_14                  string
log_archive_dest_15                  string

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
log_archive_dest_16                  string
log_archive_dest_17                  string
log_archive_dest_18                  string
log_archive_dest_19                  string
log_archive_dest_2                   string      SERVICE=PRODSTBY ASYNC VALID_F
                                                 OR=(ONLINE_LOGFILES,PRIMARY_RO
                                                 LE) DB_UNIQUE_NAME=PRODSTBY
log_archive_dest_20                  string
log_archive_dest_21                  string
log_archive_dest_22                  string
log_archive_dest_23                  string


NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
log_archive_dest_state_8             string      enable
log_archive_dest_state_9             string      enable
log_archive_duplex_dest              string
log_archive_format                   string      %t_%s_%r.dbf
log_archive_max_processes            integer     4
log_archive_min_succeed_dest         integer     1
log_archive_start                    boolean     FALSE
log_archive_trace                    integer     0
SQL>
SQL>
```

```
SQL> show parameter remote

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
remote_dependencies_mode             string      TIMESTAMP
remote_listener                      string       rac-scan:1521
remote_login_passwordfile            string      EXCLUSIVE
remote_os_authent                    boolean     FALSE
remote_os_roles                      boolean     FALSE
remote_recovery_file_dest            string
result_cache_remote_expiration       integer     0
SQL>
```

6.2 Add Standby Redo Logs (SRL)

For RAC (2 threads), SRLs created:

Thread 1: Groups 5, 6, 7
Thread 2: Groups 8, 9, 10

```
SQL> select GROUP#, THREAD#, BYTES/1024/1024 from v$log;

    GROUP#    THREAD# BYTES/1024/1024
---
         1          1             200
         2          1             200
         3          2             200
         4          2             200

SQL> select member from v$logfile;

MEMBER
--------------------------------------------------------------------------------
+DATA/PROD/ONLINELOG/group_2.263.1218045681
+RECO/PROD/ONLINELOG/group_2.258.1218045681
+DATA/PROD/ONLINELOG/group_1.262.1218045681
+RECO/PROD/ONLINELOG/group_1.257.1218045681
+DATA/PROD/ONLINELOG/group_3.269.1218047177
+RECO/PROD/ONLINELOG/group_3.259.1218047177
+DATA/PROD/ONLINELOG/group_4.270.1218047177
+RECO/PROD/ONLINELOG/group_4.260.1218047177

8 rows selected.

SQL>
SQL> alter database add standby logfile THREAD 1 group 5 ('+DATA','+RECO') size 200m;

Database altered.

SQL> alter database add standby logfile THREAD 1 group 6 ('+DATA','+RECO') size 200m;

Database altered.

SQL> alter database add standby logfile THREAD 1 group 7 ('+DATA','+RECO') size 200m;

Database altered.
```

```
SQL> alter database add standby logfile THREAD 2 group 8 ('+DATA','+RECO') size 200m;

Database altered.

SQL> alter database add standby logfile THREAD 2 group 9 ('+DATA','+RECO') size 200m;

Database altered.

SQL> alter database add standby logfile THREAD 2 group 10 ('+DATA','+RECO') size 200m;

Database altered.

SQL> select GROUP#, STATUS, TYPE, MEMBER from v$logfile;

    GROUP# STATUS  TYPE    MEMBER
---------- ------- ------- ------------------------------------------------------------
         2         ONLINE  +DATA/PROD/ONLINELOG/group_2.263.1218045681
         2         ONLINE  +RECO/PROD/ONLINELOG/group_2.258.1218045681
         1         ONLINE  +DATA/PROD/ONLINELOG/group_1.262.1218045681
         1         ONLINE  +RECO/PROD/ONLINELOG/group_1.257.1218045681
         3         ONLINE  +DATA/PROD/ONLINELOG/group_3.269.1218047177
         3         ONLINE  +RECO/PROD/ONLINELOG/group_3.259.1218047177
         4         ONLINE  +DATA/PROD/ONLINELOG/group_4.270.1218047177
         4         ONLINE  +RECO/PROD/ONLINELOG/group_4.260.1218047177
         5         STANDBY +DATA/PROD/ONLINELOG/group_5.276.1218290669
         5         STANDBY +RECO/PROD/ONLINELOG/group_5.277.1218290669
         6         STANDBY +DATA/PROD/ONLINELOG/group_6.277.1218290679
         6         STANDBY +RECO/PROD/ONLINELOG/group_6.278.1218290679
         7         STANDBY +DATA/PROD/ONLINELOG/group_7.278.1218290687
         7         STANDBY +RECO/PROD/ONLINELOG/group_7.279.1218290687
         8         STANDBY +DATA/PROD/ONLINELOG/group_8.279.1218290697
         8         STANDBY +RECO/PROD/ONLINELOG/group_8.280.1218290699
         9         STANDBY +DATA/PROD/ONLINELOG/group_9.280.1218290709
         9         STANDBY +RECO/PROD/ONLINELOG/group_9.281.1218290711
        10         STANDBY +DATA/PROD/ONLINELOG/group_10.281.1218290721
        10         STANDBY +RECO/PROD/ONLINELOG/group_10.282.1218290723

20 rows selected.


SQL> select GROUP#, THREAD#, BYTES/1024/1024 from v$log;

    GROUP#    THREAD# BYTES/1024/1024
---
         1          1             200
         2          1             200
         3          2             200
         4          2             200

4 rows selected.

SQL> select GROUP#, THREAD#, BYTES/1024/1024 from v$standby_log;

    GROUP#    THREAD# BYTES/1024/1024
---
         5          1             200
         6          1             200
         7          1             200
         8          2             200
         9          2             200
        10          2             200

6 rows selected.

SQL>
```

7. Network Configuration

7.1 Update /etc/hosts on all servers

Includes:

- Public IPs
- VIPs
- SCAN IPs
- Standby IP

Node1 :

```
[root@node1 ~]# vi /etc/hosts
[root@node1 ~]# cat /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6

# Public IPs
192.168.18.11      node1.localdomain            node1           #ens36
192.168.18.12      node2.localdomain            node2           #ens36

# Private IPs
192.168.1.11       node1-priv.localdomain       node1-priv      #ens37
192.168.1.12       node2-priv.localdomain       node2-priv      #ens37

# Virtual IPs
192.168.18.13      node1-vip.localdomain        node1-vip
192.168.18.14      node2-vip.localdomain        node2-vip

# SCAN IPs
192.168.18.15      rac-scan.localdomain         rac-scan
192.168.18.16      rac-scan.localdomain         rac-scan
192.168.18.17      rac-scan.localdomain         rac-scan


#stndby IPs
192.168.18.18      standby.localdomain           standby


[root@node1 ~]# cat /etc/hostname
node1.localdomain
[root@node1 ~]#
```

Node 2:

```
[root@node2 ~]#
[root@node2 ~]# vi /etc/hosts
[root@node2 ~]# cat /etc/hosts
```

```
[root@node2 ~]# cat /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6

# Public IPs
192.168.18.11      node1.localdomain            node1           #ens36
192.168.18.12      node2.localdomain            node2           #ens36

# Private IPs
192.168.1.11       node1-priv.localdomain       node1-priv      #ens37
192.168.1.12       node2-priv.localdomain       node2-priv      #ens37

# Virtual IPs
192.168.18.13      node1-vip.localdomain        node1-vip
192.168.18.14      node2-vip.localdomain        node2-vip

# SCAN IPs
192.168.18.15      rac-scan.localdomain         rac-scan
192.168.18.16      rac-scan.localdomain         rac-scan
192.168.18.17      rac-scan.localdomain         rac-scan


#stndby IPs
192.168.18.18      standby.localdomain           standby


[root@node2 ~]# cat /etc/hostname
node2.localdomain
[root@node2 ~]#
```

Standby :

```
[root@standby ~]# vi /etc/hosts
[root@standby ~]$ cat /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6


# Public IPs
192.168.18.11      node1.localdomain            node1           #ens36
192.168.18.12      node2.localdomain            node2           #ens36

# Private IPs
192.168.1.11       node1-priv.localdomain       node1-priv      #ens37
192.168.1.12       node2-priv.localdomain       node2-priv      #ens37

# Virtual IPs
192.168.18.13      node1-vip.localdomain        node1-vip
192.168.18.14      node2-vip.localdomain        node2-vip

# SCAN IPs
192.168.18.15      rac-scan.localdomain         rac-scan
192.168.18.16      rac-scan.localdomain         rac-scan
192.168.18.17      rac-scan.localdomain         rac-scan


#stndby IPs
192.168.18.18      standby.localdomain           standby


[root@standby ~]$ cat /etc/hostname
standby.localdomain
[root@standby ~]$
```

7.2 tnsnames.ora

Node1 :

```
[oracle@node1 ~]$ cd $ORACLE_HOME/network/admin
[oracle@node1 admin]$ pwd
/u01/app/oracle/product/19/db_1/network/admin
[oracle@node1 admin]$ ls
samples  shrept.lst  tnsnames.ora
[oracle@node1 admin]$ vi tnsnames.ora
[oracle@node1 admin]$ cat tnsnames.ora
# tnsnames.ora Network Configuration File: /u01/app/oracle/product/19/db_1/network/admin/tnsnames.ora
# Generated by Oracle configuration tools.

PROD =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = rac-scan)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = prod)
    )
  )
PRODPRMY =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = rac-scan.localdomain)(PORT = 1521))
    (CONNECT_DATA =
      (SERVICE_NAME = prod)
    )
  )

PRODSTBY =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = standby.localdomain)(PORT = 1521))
    (CONNECT_DATA =
      (SERVICE_NAME = prodstby)
    )
  )
```

Node 2:

```
[oracle@node2 ~]$ cd /u01/app/oracle/product/19/db_1/network/admin/
[oracle@node2 admin]$ ls
samples  shrept.lst  tnsnames.ora
[oracle@node2 admin]$ vi tnsnames.ora
[oracle@node2 admin]$ cat tnsnames.ora
# tnsnames.ora Network Configuration File: /u01/app/oracle/product/19/db_1/network/admin/tnsnames.ora
# Generated by Oracle configuration tools.

PROD =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = rac-scan)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = prod)
    )
  )
PRODPRMY =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = rac-scan.localdomain)(PORT = 1521))
    (CONNECT_DATA =
      (SERVICE_NAME = prod)
    )
  )

PRODSTBY =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = standby.localdomain)(PORT = 1521))
    (CONNECT_DATA =
      (SERVICE_NAME = prodstby)
    )
  )

[oracle@node2 admin]$
```

Standby :

```
[oracle@standby ~]$ cd $ORACLE_HOME/network/admin
[oracle@standby admin]$ pwd
/u01/app/oracle/product/19.0.0/dbhome_1/network/admin
[oracle@stndby admin]$ ls
samples  shrept.lst  sqlnet.ora
[oracle@standby admin]$ vi tnsnames.ora
[oracle@standby admin]$
[oracle@standby admin]$
[oracle@standby admin]$ cat tnsnames.ora

PROD =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = rac-scan)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = prod)
    )
  )
PRODPRMY =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = rac-scan.localdomain)(PORT = 1521))
    (CONNECT_DATA =
      (SERVICE_NAME = prod)
    )
  )

PRODSTBY =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = standby.localdomain)(PORT = 1521))
    (CONNECT_DATA =
      (SERVICE_NAME = prodstby)
    )
  )
```

7.3 listener.ora

Proper global names and SID entries configured.

Node1 :

```
[oracle@node1 ~]$ cd /u01/app/19/grid/network/admin/
[oracle@node1 admin]$ ls
listener2511245PM2048.bak  listener.ora.bak.node1.oracle  shrept.lst
listener.ora               samples                        sqlnet.ora
[oracle@node1 admin]$ vi listener.ora
[oracle@node1 admin]$ cat listener.ora
LISTENER=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER))))
LISTENER_SCAN3=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER_SCAN3))))
LISTENER_SCAN2=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER_SCAN2))))
LISTENER_SCAN1=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER_SCAN1))))
ASMNET1LSNR_ASM=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)(KEY=ASMNET1LSNR_ASM))))
ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER_SCAN1=ON
VALID_NODE_CHECKING_REGISTRATION_LISTENER_SCAN1=OFF
ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER_SCAN2=ON
VALID_NODE_CHECKING_REGISTRATION_LISTENER_SCAN2=OFF
ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER_SCAN3=ON
VALID_NODE_CHECKING_REGISTRATION_LISTENER_SCAN3=OFF
```

```
LISTENER_NODE1 =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = TCP)(HOST = node1-vip.localdomain)(PORT = 1521))
    )
  )

SID_LIST_LISTENER_NODE1 =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = prod)
      (ORACLE_HOME = /u01/app/oracle/product/19/db_1)
      (SID_NAME = prod1)
    )
  )

[oracle@node1 admin]$
```

Node2 :

```
[oracle@node2 ~]$ cd /u01/app/19/grid/network/admin/
[oracle@node2 admin]$ ls
listener.ora  listener.ora.bak.node2.oracle  samples  shrept.lst  sqlnet.ora
[oracle@node2 admin]$ vi listener.ora
[oracle@node2 admin]$ cat listener.ora
LISTENER_SCAN2=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER_SCAN2))))
LISTENER_SCAN3=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER_SCAN3))))
LISTENER=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER))))
LISTENER_SCAN1=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER_SCAN1))))
ASMNET1LSNR_ASM=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)(KEY=ASMNET1LSNR_ASM))))
ENABLE_GLOBAL_DYNAMIC_ENDPOINT_ASMNET1LSNR_ASM=ON
VALID_NODE_CHECKING_REGISTRATION_ASMNET1LSNR_ASM=SUBNET
ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER_SCAN1=ON
VALID_NODE_CHECKING_REGISTRATION_LISTENER_SCAN1=OFF
ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER=ON
VALID_NODE_CHECKING_REGISTRATION_LISTENER=SUBNET

LISTENER_NODE2 =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = TCP)(HOST = node2-vip.localdomain)(PORT = 1521))
    )
  )

SID_LIST_LISTENER_NODE2 =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = prod)
      (ORACLE_HOME = /u01/app/oracle/product/19/db_1)
      (SID_NAME = prod2)
    )
  )

ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER_SCAN3=ON
VALID_NODE_CHECKING_REGISTRATION_LISTENER_SCAN3=OFF
ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER_SCAN2=ON
VALID_NODE_CHECKING_REGISTRATION_LISTENER_SCAN2=OFF
[oracle@node2 admin]$
```

Standby :

```
[oracle@standby ~]$ cd /u01/app/19/grid/network/admin/
[oracle@standby admin]$ ls
listener.ora  samples  shrept.lst  sqlnet.ora
[oracle@standby admin]$ vi listener.ora
[oracle@standby admin]$ cat listener.ora
# listener.ora Network Configuration File: /u01/app/19/grid/network/admin/listener.ora
# Generated by Oracle configuration tools.

LISTENER =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = TCP)(HOST = standby.localdomain)(PORT = 1521))
      (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
    )
  )

SID_LIST_LISTENER =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = prodstby)
      (ORACLE_HOME = /u01/app/oracle/product/19/db_1)
      (SID_NAME = prod)
    )
  )

ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER=ON              # line added by Agent
VALID_NODE_CHECKING_REGISTRATION_LISTENER=ON            # line added by Agent
[oracle@standby admin]$
```

7.3 checking connectivity

Node 1 :

```
[oracle@node1 ~]$ ping node2
PING node2.localdomain (192.168.73.22) 56(84) bytes of data.
64 bytes from node2.localdomain (192.168.73.22): icmp_seq=1 ttl=64 time=0.785 ms
64 bytes from node2.localdomain (192.168.73.22): icmp_seq=2 ttl=64 time=0.507 ms
64 bytes from node2.localdomain (192.168.73.22): icmp_seq=3 ttl=64 time=0.791 ms
64 bytes from node2.localdomain (192.168.73.22): icmp_seq=4 ttl=64 time=0.642 ms
^C
--- node2.localdomain ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3072ms
rtt min/avg/max/mdev = 0.507/0.681/0.791/0.118 ms
[oracle@node1 ~]$
[oracle@node1 ~]$
[oracle@node1 ~]$ ping node2-vip
PING node2-vip.localdomain (192.168.73.24) 56(84) bytes of data.
64 bytes from node2-vip.localdomain (192.168.73.24): icmp_seq=1 ttl=64 time=0.039 ms
64 bytes from node2-vip.localdomain (192.168.73.24): icmp_seq=2 ttl=64 time=0.034 ms
64 bytes from node2-vip.localdomain (192.168.73.24): icmp_seq=3 ttl=64 time=0.038 ms
^C
--- node2-vip.localdomain ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2026ms
rtt min/avg/max/mdev = 0.034/0.037/0.039/0.002 ms
[oracle@node1 ~]$
[oracle@node1 ~]$
```

```
[oracle@node1 ~]$ ping node2-priv
PING node2-priv.localdomain (192.168.1.22) 56(84) bytes of data.
64 bytes from node2-priv.localdomain (192.168.1.22): icmp_seq=1 ttl=64 time=0.365 ms
64 bytes from node2-priv.localdomain (192.168.1.22): icmp_seq=2 ttl=64 time=0.376 ms
64 bytes from node2-priv.localdomain (192.168.1.22): icmp_seq=3 ttl=64 time=0.376 ms
^C
--- node2-priv.localdomain ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2071ms
rtt min/avg/max/mdev = 0.365/0.372/0.376/0.016 ms
[oracle@node1 ~]$
[oracle@node1 ~]$
[oracle@node1 ~]$ ping rac-scan
PING rac-scan.localdomain (192.168.73.25) 56(84) bytes of data.
64 bytes from rac-scan.localdomain (192.168.73.25): icmp_seq=1 ttl=64 time=0.025 ms
64 bytes from rac-scan.localdomain (192.168.73.25): icmp_seq=2 ttl=64 time=0.033 ms
64 bytes from rac-scan.localdomain (192.168.73.25): icmp_seq=3 ttl=64 time=0.043 ms
^C
--- rac-scan.localdomain ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2042ms
rtt min/avg/max/mdev = 0.025/0.033/0.043/0.009 ms
[oracle@node1 ~]$
[oracle@node1 ~]$

[oracle@node1 ~]$ tnsping PRODSTBY

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 27-NOV-2025 17:57:59

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Used parameter files:


Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS = (PROTOCOL = TCP)(HOST = standby.localdomain)
(PORT = 1521)) (CONNECT_DATA = (SERVICE_NAME = prodstby)))
OK (0 msec)
[oracle@node1 ~]$
[oracle@node1 ~]$
[oracle@node1 ~]$ tnsping PRODPRMY

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 27-NOV-2025 17:58:08

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Used parameter files:


Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS = (PROTOCOL = TCP)(HOST = rac-scan.localdomain)
(PORT = 1521)) (CONNECT_DATA = (SERVICE_NAME = prod)))
OK (0 msec)
[oracle@node1 ~]$
```

Node 2:

```
[oracle@node2 ~]$ ping node1
PING node1.localdomain (192.168.73.21) 56(84) bytes of data.
64 bytes from node1.localdomain (192.168.73.21): icmp_seq=1 ttl=64 time=0.586 ms
64 bytes from node1.localdomain (192.168.73.21): icmp_seq=2 ttl=64 time=0.471 ms
64 bytes from node1.localdomain (192.168.73.21): icmp_seq=3 ttl=64 time=0.400 ms
^C
```

```
--- node1.localdomain ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2081ms
rtt min/avg/max/mdev = 0.400/0.485/0.586/0.080 ms
[oracle@node2 ~]$
[oracle@node2 ~]$
[oracle@node2 ~]$ ping node1-priv
PING node1-priv.localdomain (192.168.1.21) 56(84) bytes of data.
64 bytes from node1-priv.localdomain (192.168.1.21): icmp_seq=1 ttl=64 time=0.475 ms
64 bytes from node1-priv.localdomain (192.168.1.21): icmp_seq=2 ttl=64 time=0.613 ms
64 bytes from node1-priv.localdomain (192.168.1.21): icmp_seq=3 ttl=64 time=0.532 ms
^C
--- node1-priv.localdomain ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2077ms
rtt min/avg/max/mdev = 0.475/0.540/0.613/0.056 ms
[oracle@node2 ~]$
[oracle@node2 ~]$
[oracle@node2 ~]$ ping node1-vip
PING node1-vip.localdomain (192.168.73.23) 56(84) bytes of data.
64 bytes from node1-vip.localdomain (192.168.73.23): icmp_seq=1 ttl=64 time=0.306 ms
64 bytes from node1-vip.localdomain (192.168.73.23): icmp_seq=2 ttl=64 time=0.396 ms
64 bytes from node1-vip.localdomain (192.168.73.23): icmp_seq=3 ttl=64 time=0.689 ms
^C
--- node1-vip.localdomain ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2128ms
rtt min/avg/max/mdev = 0.306/0.463/0.689/0.165 ms
[oracle@node2 ~]$
[oracle@node2 ~]$
[oracle@node2 ~]$ ping rac-scan
PING rac-scan.localdomain (192.168.73.25) 56(84) bytes of data.
64 bytes from rac-scan.localdomain (192.168.73.25): icmp_seq=1 ttl=64 time=0.532 ms
64 bytes from rac-scan.localdomain (192.168.73.25): icmp_seq=2 ttl=64 time=0.713 ms
64 bytes from rac-scan.localdomain (192.168.73.25): icmp_seq=3 ttl=64 time=0.407 ms
^C
--- rac-scan.localdomain ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2040ms
rtt min/avg/max/mdev = 0.407/0.550/0.713/0.128 ms
[oracle@node2 ~]$
[oracle@node2 ~]$
[oracle@node2 ~]$ tnsping PRODPRMY

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 27-NOV-2025 18:00:02

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Used parameter files:


Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS = (PROTOCOL = TCP)(HOST = rac-scan.localdomain)
(PORT = 1521)) (CONNECT_DATA = (SERVICE_NAME = prod)))
OK (0 msec)
[oracle@node2 ~]$
[oracle@node2 ~]$
[oracle@node2 ~]$ tnsping PRODSTBY

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 27-NOV-2025 18:00:10

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Used parameter files:


Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS = (PROTOCOL = TCP)(HOST = standby.localdomain)
(PORT = 1521)) (CONNECT_DATA = (SERVICE_NAME = prodstby)))
OK (0 msec)
[oracle@node2 ~]$
```

Standby :

```
[oracle@standby ~]$ ping node1
PING node1.localdomain (192.168.73.21) 56(84) bytes of data.
64 bytes from node1.localdomain (192.168.73.21): icmp_seq=1 ttl=64 time=0.540 ms
64 bytes from node1.localdomain (192.168.73.21): icmp_seq=2 ttl=64 time=0.693 ms
64 bytes from node1.localdomain (192.168.73.21): icmp_seq=3 ttl=64 time=0.797 ms
^C
--- node1.localdomain ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2042ms
rtt min/avg/max/mdev = 0.540/0.676/0.797/0.109 ms
[oracle@standby ~]$
[oracle@standby ~]$
[oracle@standby ~]$ ping node2
PING node2.localdomain (192.168.73.22) 56(84) bytes of data.
64 bytes from node2.localdomain (192.168.73.22): icmp_seq=1 ttl=64 time=0.465 ms
64 bytes from node2.localdomain (192.168.73.22): icmp_seq=2 ttl=64 time=0.880 ms
64 bytes from node2.localdomain (192.168.73.22): icmp_seq=3 ttl=64 time=0.811 ms
64 bytes from node2.localdomain (192.168.73.22): icmp_seq=4 ttl=64 time=0.578 ms
^C
--- node2.localdomain ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3094ms
rtt min/avg/max/mdev = 0.465/0.683/0.880/0.170 ms
[oracle@standby ~]$
[oracle@standby ~]$
[oracle@standby ~]$ ping node2-vip
PING node2-vip.localdomain (192.168.73.24) 56(84) bytes of data.
64 bytes from node2-vip.localdomain (192.168.73.24): icmp_seq=1 ttl=64 time=0.966 ms
64 bytes from node2-vip.localdomain (192.168.73.24): icmp_seq=2 ttl=64 time=0.468 ms
64 bytes from node2-vip.localdomain (192.168.73.24): icmp_seq=3 ttl=64 time=0.622 ms
64 bytes from node2-vip.localdomain (192.168.73.24): icmp_seq=4 ttl=64 time=0.664 ms
64 bytes from node2-vip.localdomain (192.168.73.24): icmp_seq=5 ttl=64 time=0.524 ms
^C
--- node2-vip.localdomain ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4095ms
rtt min/avg/max/mdev = 0.468/0.648/0.966/0.176 ms
[oracle@standby ~]$
[oracle@standby ~]$
[oracle@standby ~]$ ping node1-vip
PING node1-vip.localdomain (192.168.73.23) 56(84) bytes of data.
64 bytes from node1-vip.localdomain (192.168.73.23): icmp_seq=1 ttl=64 time=0.539 ms
64 bytes from node1-vip.localdomain (192.168.73.23): icmp_seq=2 ttl=64 time=0.456 ms
64 bytes from node1-vip.localdomain (192.168.73.23): icmp_seq=3 ttl=64 time=0.751 ms
64 bytes from node1-vip.localdomain (192.168.73.23): icmp_seq=4 ttl=64 time=0.448 ms
^C
--- node1-vip.localdomain ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3057ms
rtt min/avg/max/mdev = 0.448/0.548/0.751/0.124 ms
[oracle@standby ~]$
[oracle@standby ~]$ tnsping PRODPRMY

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 27-NOV-2025 21:04:03

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Used parameter files:


Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS = (PROTOCOL = TCP)(HOST = rac-scan.localdomain)
(PORT = 1521)) (CONNECT_DATA = (SERVICE_NAME = prod)))
OK (0 msec)
[oracle@standby ~]$
[oracle@standby ~]$
```

```
[oracle@standby ~]$ tnsping PRODSTBY

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 27-NOV-2025 21:04:46

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Used parameter files:


Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS = (PROTOCOL = TCP)(HOST = standby.localdomain)
(PORT = 1521)) (CONNECT_DATA = (SERVICE_NAME = prodstby)))
OK (0 msec)
[oracle@standby ~]$
```

---

8. Password File & Initialization File Setup

Copy Password File from ASM and ORACLE_HOME/dbs

Create PFILE and Copy

```
[oracle@node1 ~]$ . oraenv
ORACLE_SID = [prod1] ? +ASM1
The Oracle base remains unchanged with value /u01/app/oracle
[oracle@node1 ~]$
[oracle@node1 ~]$
[oracle@node1 ~]$ asmcmd
ASMCMD>
ASMCMD> pwd
+
ASMCMD> ls
DATA/
OCRVD/
RECO/
ASMCMD> cd DATA
ASMCMD> ls
PROD/
ASMCMD> cd PROD
ASMCMD> ls
44598EB20DEE450EE0630B12A8C0935A/
4459E1E26D9D7828E0630B12A8C0116B/
86B637B62FE07A65E053F706E80A27CA/
CONTROLFILE/
DATAFILE/
ONLINELOG/
PARAMETERFILE/
PASSWORD/
TEMPFILE/
ASMCMD> cd PASSWORD
ASMCMD> ls
pwdprod.256.1218045447
ASMCMD>
ASMCMD> pwcopy pwdprod.256.1218045447 /tmp/orapwprod
copying +DATA/PROD/PASSWORD/pwdprod.256.1218045447 -> /tmp/orapwprod
ASMCMD>
ASMCMD> exit
```

```
[oracle@node1 ~]$ scp /tmp/orapwprod oracle@192.168.18.18:/tmp/
The authenticity of host '192.168.18.18 (192.168.18.18)' can't be established.
ECDSA key fingerprint is SHA256:LkCcu9WDbtiBbvsPuBxdu3c+NK4fngM+N6+3WF6gXN8.
ECDSA key fingerprint is MD5:b0:a2:68:60:15:3d:0f:93:87:c0:5a:cb:13:95:2a:06.
Are you sure you want to continue connecting (yes/no)? yes

Warning: Permanently added '192.168.18.18' (ECDSA) to the list of known hosts.
oracle@192.168.18.18's password:
orapwprod                                     100% 2048   291.9KB/s   00:00
[oracle@node1 ~]$
[oracle@node1 ~]$ . oraenv
ORACLE_SID = [prod1] ? prod1
The Oracle base remains unchanged with value /u01/app/oracle
[oracle@node1 ~]$
[oracle@node1 ~]$
[oracle@node1 ~]$ cd $ORACLE_HOME/dbs
[oracle@node1 dbs]$ ls
hc_prod1.dat  id_prod1.dat  init.ora
[oracle@node1 dbs]$
[oracle@node1 dbs]$
[oracle@node1 dbs]$ orapwd file=$ORACLE_HOME/dbs/orapwprod password='Welcome$123' entries=5
[oracle@node1 dbs]$
[oracle@node1 dbs]$ ls
hc_prod1.dat  id_prod1.dat  init.ora  orapwprod
[oracle@node1 dbs]$
[oracle@node1 dbs]$
[oracle@node1 dbs]$
[oracle@node1 dbs]$ scp orapwprod oracle@192.168.18.18:/u01/app/oracle/product/19/db_1/dbs/
oracle@192.168.18.18's password:
orapwprod                                     100% 6144   961.6KB/s   00:00
[oracle@node1 dbs]$
[oracle@node1 dbs]$ !sql
sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Nov 27 14:28:06 2025
Version 19.3.0.0.0

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Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> create pfile from spfile;

File created.

SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
[oracle@node1 dbs]$ ls

 hc_prod1.dat  id_prod1.dat  init.ora  initprod1.ora  orapwprod

 [oracle@node1 dbs]$
 [oracle@node1 dbs]$
 [oracle@node1 dbs]$ scp initprod1.ora oracle@192.168.18.18:/u01/app/oracle/product/19/db_1/dbs/initprod.ora
 oracle@192.168.18.18's password:
 initprod1.ora                                 100% 2259   541.6KB/s   00:00
 [oracle@node1 dbs]$
```

9. Creating the Standby Database

9.1 Start Standby NOMOUNT

```
[oracle@standby ~]$ export ORACLE_SID=prod
[oracle@standby ~]$ export ORACLE_HOME=/u01/app/oracle/product/19/db_1
[oracle@standby ~]$ export PATH=$ORACLE_HOME/bin:$PATH
[oracle@standby ~]$
[oracle@standby ~]$ cd $ORACLE_HOME/dbs
[oracle@standby dbs]$
[oracle@standby dbs]$ pwd
/u01/app/oracle/product/19/db_1/dbs
[oracle@standby dbs]$
[oracle@standby dbs]$ ls
init.ora  initprod.ora  orapwprod
[oracle@standby dbs]$ vi initprod.ora
[oracle@standby dbs]$ cat initprod.ora

*.audit_file_dest='/u01/app/oracle/admin/prod/adump'
*.audit_trail='db'
*.cluster_database=false
*.compatible='19.0.0'
*.control_files='+DATA/PROD/CONTROLFILE/current.261.1218045679','+RECO/PROD/CONTROLFILE/current.256.1218045679'
*.db_block_size=8192
*.db_create_file_dest='+DATA'
*.db_name='prod'
*.db_unique_name='prodstby'
*.db_recovery_file_dest='+RECO'
*.db_recovery_file_dest_size=19732m
*.diagnostic_dest='/u01/app/oracle'
*.dispatchers='(PROTOCOL=TCP) (SERVICE=prodXDB)'
*.enable_pluggable_database=true
*.fal_client='PRODSTBY'
*.fal_server='PROD'
family:dw_helper.instance_mode='read-only'
*.local_listener='-oraagent-dummy-'
*.log_archive_config='DG_CONFIG=(PRODSTBY,PROD)'
*.log_archive_dest_1='LOCATION=USE_DB_RECOVERY_FILE_DEST VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=PRODSTBY'
*.log_archive_dest_2='SERVICE=PROD ASYNC VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=PROD'
*.log_archive_dest_state_1='ENABLE'
*.log_archive_dest_state_2='ENABLE'
*.log_archive_format='%t_%s_%r.dbf'
*.nls_language='AMERICAN'
*.nls_territory='AMERICA'
*.open_cursors=300
*.pga_aggregate_target=966m
*.processes=300
*.remote_login_passwordfile='EXCLUSIVE'
*.sga_target=2896m
*.standby_file_management='AUTO'
undo_tablespace='UNDOTBS1'

[oracle@standby dbs]$
[oracle@standby dbs]$ mkdir -p /u01/app/oracle/admin/prod/adump
[oracle@standby dbs]$

[oracle@standby dbs]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Nov 27 14:39:09 2025
Version 19.3.0.0.0

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Connected to an idle instance.

SQL> startup nomount pfile='/u01/app/oracle/product/19/db_1/dbs/initprod.ora';
ORACLE instance started.
```

```
Total System Global Area 3036673240 bytes
Fixed Size                  9139416 bytes
Variable Size             620756992 bytes
Database Buffers         2399141888 bytes
Redo Buffers                7634944 bytes
SQL>
SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
[oracle@standby dbs]$
```

9.2 RMAN Duplicate Command

RMAN will:

✓ Create Controlfile
✓ Create Datafiles (OMF)
✓ Restore DB
✓ Recover DB
✓ Mount Standby

```
[oracle@standby ~]$ rman target sys@PRODPRMY auxiliary sys@PRODSTBY

Recovery Manager: Release 19.0.0.0.0 - Production on Thu Nov 27 14:47:10 2025
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle and/or its affiliates.  All rights reserved.

target database Password:
connected to target database: PROD (DBID=629884015)
auxiliary database Password:
connected to auxiliary database: PROD (not mounted)

RMAN>

RMAN> Duplicate target database for standby from active database dorecover nofilenamecheck;

Starting Duplicate Db at 27-NOV-25
using target database control file instead of recovery catalog
allocated channel: ORA_AUX_DISK_1
channel ORA_AUX_DISK_1: SID=258 device type=DISK
current log archived

contents of Memory Script:
{
   backup as copy reuse
   passwordfile auxiliary format  '/u01/app/oracle/product/19/db_1/dbs/orapwprod'    ;
}
executing Memory Script
```

```
Starting backup at 27-NOV-25
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=295 instance=prod2 device type=DISK
Finished backup at 27-NOV-25
duplicating Online logs to Oracle Managed File (OMF) location
duplicating Datafiles to Oracle Managed File (OMF) location

contents of Memory Script:
{
   sql clone "create spfile from memory";
   shutdown clone immediate;
   startup clone nomount;
   restore clone from service  'PRODPRMY' standby controlfile;
}
executing Memory Script

sql statement: create spfile from memory

Oracle instance shut down

connected to auxiliary database (not started)
Oracle instance started

Total System Global Area    3036673240 bytes

Fixed Size                     9139416 bytes
Variable Size                620756992 bytes
Database Buffers            2399141888 bytes
Redo Buffers                   7634944 bytes

Starting restore at 27-NOV-25
allocated channel: ORA_AUX_DISK_1
channel ORA_AUX_DISK_1: SID=259 device type=DISK

channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service PRODPRMY
channel ORA_AUX_DISK_1: restoring control file
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:05
output file name=+DATA/PRODSTBY/CONTROLFILE/current.257.1218293335
output file name=+RECO/PRODSTBY/CONTROLFILE/current.257.1218293337
Finished restore at 27-NOV-25

contents of Memory Script:
{
   sql clone 'alter database mount standby database';
}
executing Memory Script

sql statement: alter database mount standby database

contents of Memory Script:
{
   set newname for clone tempfile  1 to new;
   set newname for clone tempfile  2 to new;
   set newname for clone tempfile  3 to new;
   switch clone tempfile all;
   set newname for clone datafile  1 to new;
   set newname for clone datafile  3 to new;
   set newname for clone datafile  4 to new;
   set newname for clone datafile  5 to new;
   set newname for clone datafile  6 to new;
   set newname for clone datafile  7 to new;
   set newname for clone datafile  9 to new;
```

```
   set newname for clone datafile  10 to new;
   set newname for clone datafile  11 to new;
   set newname for clone datafile  12 to new;
   restore
   from  nonsparse   from service
 'PRODPRMY'   clone database
   ;
   sql 'alter system archive log current';
}
executing Memory Script

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

renamed tempfile 1 to +DATA in control file
renamed tempfile 2 to +DATA in control file
renamed tempfile 3 to +DATA in control file

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

Starting restore at 27-NOV-25
using channel ORA_AUX_DISK_1


channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service PRODPRMY
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00001 to +DATA
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:15
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service PRODPRMY
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00003 to +DATA
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:08
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service PRODPRMY
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
```

```
Finished restore at 27-NOV-25

sql statement: alter system archive log current
current log archived

contents of Memory Script:
{
   restore clone force from service  'PRODPRMY'
            archivelog from scn  2517504;
   switch clone datafile all;
}
executing Memory Script

Starting restore at 27-NOV-25
using channel ORA_AUX_DISK_1

channel ORA_AUX_DISK_1: starting archived log restore to default destination
channel ORA_AUX_DISK_1: using network backup set from service PRODPRMY
channel ORA_AUX_DISK_1: restoring archived log
archived log thread=1 sequence=14
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
channel ORA_AUX_DISK_1: starting archived log restore to default destination
channel ORA_AUX_DISK_1: using network backup set from service PRODPRMY
channel ORA_AUX_DISK_1: restoring archived log
archived log thread=1 sequence=15
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:02
channel ORA_AUX_DISK_1: starting archived log restore to default destination
channel ORA_AUX_DISK_1: using network backup set from service PRODPRMY
channel ORA_AUX_DISK_1: restoring archived log
archived log thread=1 sequence=16
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
channel ORA_AUX_DISK_1: starting archived log restore to default destination
channel ORA_AUX_DISK_1: using network backup set from service PRODPRMY
channel ORA_AUX_DISK_1: restoring archived log
archived log thread=2 sequence=9
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:02
channel ORA_AUX_DISK_1: starting archived log restore to default destination
channel ORA_AUX_DISK_1: using network backup set from service PRODPRMY
channel ORA_AUX_DISK_1: restoring archived log
archived log thread=2 sequence=10
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
Finished restore at 27-NOV-25

datafile 1 switched to datafile copy
input datafile copy RECID=14 STAMP=1218293425 file name=+DATA/PRODSTBY/DATAFILE/system.258.1218293347
datafile 3 switched to datafile copy
input datafile copy RECID=15 STAMP=1218293425 file name=+DATA/PRODSTBY/DATAFILE/sysaux.259.1218293363
datafile 4 switched to datafile copy
input datafile copy RECID=16 STAMP=1218293425 file name=+DATA/PRODSTBY/DATAFILE/undotbs1.260.1218293337
datafile 5 switched to datafile copy
input datafile copy RECID=17 STAMP=1218293425 file name=+DATA/PRODSTBY/44598EB20DEE450EE0630B12A8C0935
datafile 6 switched to datafile copy
input datafile copy RECID=18 STAMP=1218293425 file name=+DATA/PRODSTBY/44598EB20DEE450EE0630B12A8C0935
datafile 7 switched to datafile copy
input datafile copy RECID=19 STAMP=1218293425 file name=+DATA/PRODSTBY/DATAFILE/users.263.1218293391
datafile 9 switched to datafile copy
input datafile copy RECID=20 STAMP=1218293425 file name=+DATA/PRODSTBY/DATAFILE/undotbs2.264.1218293339
datafile 10 switched to datafile copy
input datafile copy RECID=21 STAMP=1218293425 file name=+DATA/PRODSTBY/4459E1E26D9D7828E0630B12A8C0116
datafile 11 switched to datafile copy
input datafile copy RECID=22 STAMP=1218293425 file name=+DATA/PRODSTBY/4459E1E26D9D7828E0630B12A8C0116
datafile 12 switched to datafile copy
input datafile copy RECID=23 STAMP=1218293425 file name=+DATA/PRODSTBY/4459E1E26D9D7828E0630B12A8C0116
```

```
contents of Memory Script:
{
   set until scn  2518050;
   recover
   standby
   clone database
    delete archivelog
   ;
}
executing Memory Script

executing command: SET until clause

Starting recover at 27-NOV-25
using channel ORA_AUX_DISK_1

starting media recovery


archived log for thread 1 with sequence 15 is already on disk as file +RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thr
archived log for thread 1 with sequence 16 is already on disk as file +RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thr
archived log for thread 2 with sequence 9 is already on disk as file +RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thre
archived log for thread 2 with sequence 10 is already on disk as file +RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thr
archived log file name=+RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thread_1_seq_15.259.1218293419 thread=1 sequence=1
archived log file name=+RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thread_2_seq_9.261.1218293423 thread=2 sequence=9
archived log file name=+RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thread_1_seq_16.260.1218293421 thread=1 sequence=1
archived log file name=+RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thread_2_seq_10.262.1218293423 thread=2 sequence=1
media recovery complete, elapsed time: 00:00:02
Finished recover at 27-NOV-25

contents of Memory Script:
{
   delete clone force archivelog all;
}
executing Memory Script

released channel: ORA_DISK_1
released channel: ORA_AUX_DISK_1
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=295 instance=prod2 device type=DISK
deleted archived log
archived log file name=+RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thread_1_seq_14.258.1218293417 RECID=1
deleted archived log
archived log file name=+RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thread_1_seq_15.259.1218293419 RECID=2
deleted archived log
archived log file name=+RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thread_1_seq_16.260.1218293421 RECID=3
deleted archived log
archived log file name=+RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thread_2_seq_9.261.1218293423 RECID=4
deleted archived log
archived log file name=+RECO/PRODSTBY/ARCHIVELOG/2025_11_27/thread_2_seq_10.262.1218293423 RECID=5
Deleted 5 objects

Finished Duplicate Db at 27-NOV-25

RMAN> exit

Recovery Manager complete.
```

10. Start Redo Apply

```
[oracle@standby ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Nov 27 15:20:19 2025
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle.  All rights reserved.


Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> select status from v$instance;

STATUS
------------
MOUNTED

SQL> alter database recover managed standby database disconnect from session;

Database altered.

SQL>
SQL> show parameter name

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
cdb_cluster_name                     string
cell_offloadgroup_name               string
db_file_name_convert                 string
db_name                              string      prod
db_unique_name                       string      prodstby
global_names                         boolean     FALSE
instance_name                        string      prod
lock_name_space                      string
log_file_name_convert                string
pdb_file_name_convert                string
processor_group_name                 string

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
service_names                        string      prodstby
SQL>
```

---

11. Verification

11.1 PRODUCTION side: Primary RAC (prod)

```
SQL> SELECT name, open_mode, database_role, db_unique_name FROM v$database;

NAME      OPEN_MODE            DATABASE_ROLE    DB_UNIQUE_NAME
--------- -------------------- ---------------- ------------------------------
PROD      READ WRITE           PRIMARY          PROD

SQL>
```

```
SQL> select thread#,max(sequence#) from v$archived_log group by thread#;

     THREAD# MAX(SEQUENCE#)
---------- --------------
         1             16
         2             13
```

11.2 DR side:standby standalone (prod)

```
SQL> SELECT name, open_mode, database_role, db_unique_name FROM v$database;

NAME      OPEN_MODE            DATABASE_ROLE    DB_UNIQUE_NAME
--------- -------------------- ---------------- ------------------------------
PROD      MOUNTED              PHYSICAL STANDBY prodstby

SQL>

SQL> select thread#,max(sequence#) from v$archived_log group by thread#;

     THREAD# MAX(SEQUENCE#)
---------- --------------
         1             16
         2             13
```

11.3 Check Lag

```
SQL> SELECT name, value, unit FROM v$dataguard_stats
WHERE name IN ('transport lag','apply lag','apply rate');

NAME                           VALUE                                                                        UNIT
------------------------------ ---------------------------------------------------------------------------- ------------------------------
transport lag                                                                                               day(2) to second(0) interval
apply lag                                                                                                   day(2) to second(0) interval

SQL> SELECT PROCESS, STATUS, THREAD#, SEQUENCE# FROM v$managed_standby;


PROCESS   STATUS          THREAD#  SEQUENCE#
--------- ------------ ---------- ----------
ARCH      CLOSING               2         11
DGRD      ALLOCATED             0          0
DGRD      ALLOCATED             0          0
ARCH      CLOSING               2         13
ARCH      CONNECTED             0          0
ARCH      CONNECTED             0          0
RFS       IDLE                  1          0
RFS       IDLE                  1         17
RFS       IDLE                  0          0
RFS       IDLE                  0          0
RFS       IDLE                  0          0

PROCESS   STATUS          THREAD#  SEQUENCE#
--------- ------------ ---------- ----------
RFS       IDLE                  0          0
RFS       IDLE                  2         14
RFS       IDLE                  2          0
MRP0      APPLYING_LOG          2         14

15 rows selected.
```

All values confirmed synchronized.

11.4 Real-Time Log Apply Verification by Manual Log Switch

To verify that the physical standby database is receiving redo in real time and applying it correctly, we manually force a log switch on the primary database. This sends a new archived log to the standby, and the Managed Recovery Process (MRP) should immediately begin applying it.

Primary

>Force Redo Log Switch on Primary

>Check Latest Archive Log Sequence on Primary

```
SQL> alter system switch logfile;

System altered.

SQL> /

System altered.

SQL> /

System altered.

SQL> /

System altered.

SQL> /

System altered.

SQL> select thread#,max(sequence#) from v$archived_log group by thread#;

   THREAD# MAX(SEQUENCE#)
---------- --------------
         1             21
         2             14

SQL>
```

Standby

>Verify Redo Reception on Standby

```
SQL> select thread#,max(sequence#) from v$archived_log group by thread#;

   THREAD# MAX(SEQUENCE#)
---------- --------------
         1             21
         2             14
```

12. Data Guard Broker Configuration

12.1 Enable Broker on Both DBs

Primary

```
SQL> show parameter dg_broker

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
dg_broker_config_file1               string      /u01/app/oracle/product/19/db_
                                                 1/dbs/dr1prod.dat
dg_broker_config_file2               string      /u01/app/oracle/product/19/db_
                                                 1/dbs/dr2prod.dat
dg_broker_start                      boolean     FALSE
SQL>
SQL>
SQL>
SQL>
SQL> ALTER SYSTEM SET dg_broker_start = TRUE SCOPE=BOTH SID='*';

System altered.

SQL>
SQL>
SQL> show parameter dg_broker

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
dg_broker_config_file1               string      /u01/app/oracle/product/19/db_
                                                 1/dbs/dr1prod.dat
dg_broker_config_file2               string      /u01/app/oracle/product/19/db_
                                                 1/dbs/dr2prod.dat
dg_broker_start                      boolean     TRUE
SQL>
```

Standby

```
SQL> show parameter dg_broker

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
dg_broker_config_file1               string      /u01/app/oracle/product/19/db_
                                                 1/dbs/dr1prodstby.dat
dg_broker_config_file2               string      /u01/app/oracle/product/19/db_
                                                 1/dbs/dr2prodstby.dat
dg_broker_start                      boolean     FALSE
SQL>
SQL>
SQL> ALTER SYSTEM SET dg_broker_start = TRUE SCOPE=BOTH SID='*';

System altered.

SQL> show parameter dg_broker

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
dg_broker_config_file1               string      /u01/app/oracle/product/19/db_
                                                 1/dbs/dr1prodstby.dat
dg_broker_config_file2               string      /u01/app/oracle/product/19/db_
                                                 1/dbs/dr2prodstby.dat
dg_broker_start                      boolean     TRUE
SQL>
```

12.2 Removing MRP Before Data Guard Broker Configuration

Before configuring Data Guard Broker, it is important to **stop the Managed Recovery Process (MRP)** on the standby database.
MRP is responsible for applying redo logs, but when active, it can interfere with Broker operations such as:

- enabling the Data Guard configuration
- adding the standby database to the broker
- performing switchover/failover
- reading the correct database state information

Data Guard Broker automatically manages redo apply, so it must take full control.
Therefore, **MRP must be stopped manually** before enabling DG Broker.

```
SQL> ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;

Database altered.

SQL> SELECT PROCESS, STATUS, THREAD#, SEQUENCE# FROM v$managed_standby;

PROCESS   STATUS          THREAD#  SEQUENCE#
--------- ------------ ---------- ----------
ARCH      CLOSING               2         17
DGRD      ALLOCATED             0          0
DGRD      ALLOCATED             0          0
ARCH      CLOSING               1         27
ARCH      CLOSING               2         18
ARCH      CLOSING               1         28
RFS       IDLE                  2          0
RFS       IDLE                  1          0
RFS       RECEIVING             2         19
RFS       IDLE                  1         29

10 rows selected.
```

12.3 Create DG Configuration

```
[oracle@node1 ~]$ dgmgrl
DGMGRL for Linux: Release 19.0.0.0.0 - Production on Thu Nov 27 15:58:43 2025
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle and/or its affiliates.  All rights reserved.

Welcome to DGMGRL, type "help" for information.
DGMGRL> connect sys@PROD
Password:
Connected to "prod"
Connected as SYSDBA.
```

```
DGMGRL> CREATE CONFIGURATION 'PROD_DG' AS PRIMARY DATABASE IS 'PROD' CONNECT IDENTIFIER IS PROD;
Configuration "PROD_DG" created with primary database "PROD"
DGMGRL> ADD DATABASE 'PRODSTBY' AS CONNECT IDENTIFIER IS PRODSTBY MAINTAINED AS PHYSICAL;
Error: ORA-16698: member has a LOG_ARCHIVE_DEST_n parameter with SERVICE attribute set

Failed.
DGMGRL>
```

NOTE

When using Broker, manual redo routes (SERVICE=…) must be removed.

Broker will automatically create:

LOG_ARCHIVE_DEST_2 on Primary to Standby

LOG_ARCHIVE_DEST_2 on Standby back to Primary

This allows:

Automatic failover

Switchover

Health checks

Automatic redo routing after role transitions

If manual SERVICE=… exists, the broker does NOT take control — this creates the ORA-16698 conflict.

PRIMARY :

```
[oracle@node1 ~]$ !sql
sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Nov 27 16:02:52 2025
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle.  All rights reserved.


Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> alter system set LOG_ARCHIVE_DEST_2='' scope=both sid='*';

System altered.
```

```
SQL>
SQL> alter system set dg_broker_start=false scope=both sid='*';

System altered.

SQL> alter system set dg_broker_start=true scope=both sid='*';

System altered.

SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
[oracle@node1 ~]$
```

Standby :

```
[oracle@standby ~]$ !sql
sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Nov 27 16:03:12 2025
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle.  All rights reserved.


Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> alter system set LOG_ARCHIVE_DEST_2='' scope=both sid='*';

System altered.

SQL>
SQL> alter system set dg_broker_start=false scope=both sid='*';

System altered.

SQL> alter system set dg_broker_start=true scope=both sid='*';

System altered.
```

Begin Data Guard Broker Configuration

```
DGMGRL> connect sys@PRODPRMY
Password:
Connected to "prod"
Connected as SYSDBA.
DGMGRL> CREATE CONFIGURATION 'PROD_DG' AS PRIMARY DATABASE IS 'PROD' CONNECT IDENTIFIER IS PRODPRMY;
Configuration "PROD_DG" created with primary database "PROD"
DGMGRL> ADD DATABASE 'PRODSTBY' AS CONNECT IDENTIFIER IS PRODSTBY MAINTAINED AS PHYSICAL;
Database "PRODSTBY" added
DGMGRL> ENABLE CONFIGURATION;
Enabled.
```

12.4 Verify

```
DGMGRL> SHOW CONFIGURATION;

Configuration - PROD_DG

  Protection Mode: MaxPerformance
  Members:
  PROD     - Primary database
    PRODSTBY - Physical standby database

Fast-Start Failover:  Disabled

Configuration Status:
SUCCESS   (status updated 47 seconds ago)

DGMGRL>
DGMGRL>
DGMGRL> SHOW DATABASE PROD;

Database - PROD

  Role:               PRIMARY
  Intended State:     TRANSPORT-ON
  Instance(s):
    prod1
    prod2

Database Status:
SUCCESS

DGMGRL> SHOW DATABASE PRODSTBY;

Database - PRODSTBY

  Role:               PHYSICAL STANDBY
  Intended State:     APPLY-ON
  Transport Lag:      0 seconds (computed 0 seconds ago)
  Apply Lag:          0 seconds (computed 0 seconds ago)
  Average Apply Rate: 221.00 KByte/s
  Real Time Query:    OFF
  Instance(s):
    prod

Database Status:
SUCCESS

DGMGRL>
```

13. Switchover Operation in Data Guard Broker

A **Switchover** is a planned role reversal between the **Primary** and the **Standby** database, performed when both databases are healthy.
It is a *zero data loss* operation and is typically done for:

- maintenance activities on the primary
- testing the disaster recovery (DR) site
- validating Data Guard configuration
- planned failover simulations

During a switchover:

- The **Primary** becomes a **Standby**
- The **Standby** becomes the new **Primary**
- Redo transport and apply automatically reverse directions
- Data Guard Broker handles all internal steps

Switchover is **safe**, **fast**, and **fully reversible** as long as both databases are synchronized.

13.1 Prerequisites Before Performing Switchover

Ensure the following:

1. Data Guard Broker Configuration Status = SUCCESS

```
DGMGRL>
DGMGRL> SHOW CONFIGURATION;

Configuration - PROD_DG

  Protection Mode: MaxPerformance
  Members:
  PROD     - Primary database
    PRODSTBY - Physical standby database

Fast-Start Failover:  Disabled

Configuration Status:
SUCCESS   (status updated 35 seconds ago)

DGMGRL> show database prod

Database - PROD

  Role:               PRIMARY
  Intended State:     TRANSPORT-ON
  Instance(s):
    prod1
    prod2

Database Status:
SUCCESS
```

2. Both databases have no apply/transport lag

```
DGMGRL> validate database prod

  Database Role:    Primary database

  Ready for Switchover:  Yes

  Flashback Database Status:
    PROD:  Off

  Managed by Clusterware:
    PROD:  YES

DGMGRL> validate database prodstby

  Database Role:     Physical standby database
  Primary Database:  PROD

  Ready for Switchover:  Yes
  Ready for Failover:    Yes (Primary Running)

  Flashback Database Status:
    PROD    :  Off
    PRODSTBY:  Off

  Capacity Information:
    Database  Instances        Threads
    PROD      2                2
    PRODSTBY  1                2
    Warning: the target standby has fewer instances than the
    primary database, this may impact application performance

  Managed by Clusterware:
    PROD    :  YES
    PRODSTBY:  NO

  Log Files Cleared:
    PROD Standby Redo Log Files:      Cleared
    PRODSTBY Online Redo Log Files:   Not Cleared
    PRODSTBY Standby Redo Log Files:  Available

DGMGRL>
```

13.2 Execute the Switchover Using DGMGRL

Step 1: Login to DGMGRL

```
[oracle@node1 ~]$ dgmgrl sys@PRODPRMY
DGMGRL for Linux: Release 19.0.0.0.0 - Production on Sun Dec 7 14:59:02 2025
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle and/or its affiliates.  All rights reserved.

Welcome to DGMGRL, type "help" for information.
Password:
Connected to "prod"
Connected as SYSDBA.
DGMGRL>
```

Step 2: Perform the Switchover

During switchover:

- Broker stops redo apply
- Converts the standby database to primary
- Converts the original primary to standby
- Starts redo apply on the new standby

```
DGMGRL>
DGMGRL> switchover to PRODSTBY;
Performing switchover NOW, please wait...
Operation requires a connection to database "PRODSTBY"
Connecting ...
Connected to "prodstby"
Connected as SYSDBA.
New primary database "PRODSTBY" is opening...
Oracle Clusterware is restarting database "PROD" ...
Connected to "prod"
Connected to "prod"
Switchover succeeded, new primary is "prodstby"
DGMGRL>
```

13.3 Post-Switchover Verification

Check configuration status

```
DGMGRL> show configuration;

Configuration - PROD_DG

  Protection Mode: MaxPerformance
  Members:
  PRODSTBY - Primary database
    PROD     - Physical standby database

Fast-Start Failover:  Disabled

Configuration Status:
SUCCESS   (status updated 108 seconds ago)

DGMGRL>
```

The Data Guard Broker configuration (PROD_DG) shows STATUS = **SUCCESS**, confirming that both the primary database (PRODSTBY) and the physical standby database (PROD) are healthy, synchronized, and communicating properly. The configuration is operating in Maximum Performance mode, with no transport or apply issues. This validates that the **switchover was successful** and the Data Guard environment is functioning normally.

Verify new database roles

On new primary (PRODSTBY):

```
[oracle@standby ~]$ !sql
sqlplus / as sysdba;

SQL*Plus: Release 19.0.0.0.0 - Production on Sun Dec 7 14:29:53 2025
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle.  All rights reserved.


Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> SELECT name, open_mode, database_role, db_unique_name FROM v$database;

NAME      OPEN_MODE            DATABASE_ROLE    DB_UNIQUE_NAME
--------- -------------------- ---------------- ------------------------------
PROD      READ WRITE           PRIMARY          prodstby

SQL>
```

On new standby (PROD):

```
[oracle@node1 ~]$ !sql
sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sun Dec 7 14:30:17 2025
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle.  All rights reserved.


Connected to:
Oracle Database 19c Enterprise Edition Release 19.0[oracle@standby ~]$ !sql
sqlplus / as sysdba;

SQL*Plus: Release 19.0.0.0.0 - Production on Sun Dec 7 14:29:53 2025
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle.  All rights reserved.


Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> SELECT name, open_mode, database_role, db_unique_name FROM v$database;

NAME      OPEN_MODE            DATABASE_ROLE    DB_UNIQUE_NAME
--------- -------------------- ---------------- ------------------------------
PROD      MOUNTED              PHYSICAL STANDBY prod

SQL>
```

13.4 Benefits of Performing a Switchover

- Zero data loss (safe operation)
- Validates HA/DR readiness
- Confirms Data Guard health
- Used for patching or maintenance
- Ensures both sites are operational and synchronized

---

14. Final Status Summary

Item	Status
Primary DB	PROD (RAC), OPEN
Standby DB	PRODSTBY (Standalone), APPLYING LOGS
Data Guard	ACTIVE, SYNCHRONIZED
Log Transport	WORKING
Broker	ENABLED & HEALTHY
PDBs	Successfully replicated

---

15. Conclusion

This implementation successfully establishes a **robust disaster recovery environment** using Oracle Data Guard between a highly available **2-node RAC primary** and a **standalone ASM-based physical standby**.

With **Data Guard Broker**, database role transitions become easier, faster, and automated, ensuring maximum uptime and simplified administration.

This setup aligns with best practices for:

- **High Availability (HA)**
- **Disaster Recovery (DR)**
- **Minimal downtime architecture**
- **Enterprise-level data protection**